

Analysis of Memristors and Resistive Switching in Computation and Neuromorphic Circuit Design

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Abstract—The Von Neumann architecture faces a critical ‘Memory Wall’ when processing data-intensive neural networks. This bottleneck can be addressed by the memristor, a non-linear circuit element enabling Compute-in-Memory. This paper reviews memristive computing and investigates circuit-level implementations using SPICE simulations[1] as well as large-scale simulations with Julia. We compare 0T1R and 1T1R architectures, analyzing sneak-path leakage and scalability. Finally, we examine non-ideal effects such as IR drop, device nonlinearity, and peripheral overhead that constrain large-scale neuromorphic cores. We demonstrate the framework on single-device and crossbar-level benchmarks, showing close agreement between Julia and SPICE simulations. The proposed approach enables scalable, hardware-aware neuromorphic system design and provides a pathway toward next-generation co-design workflows for memristive computing.

Index Terms—IEEE, Neuromorphic, Memristor, Computation, Spice, Simulation

I. INTRODUCTION

THE trajectory of computing faces fundamental physical and architectural limits. For decades, the von Neumann architecture, characterized by the separation of CPU and memory, has been the bedrock of modern systems. However, data-intensive applications like deep learning have exposed a critical “von Neumann bottleneck,” where energy and latency costs of data movement dwarf computation itself. This “Memory Wall,” combined with the “Power Wall” of scaling CMOS, necessitates a paradigm shift.

To circumvent these barriers, the field has pivoted toward “Compute-in-Memory” (CIM) and neuromorphic architectures, which emulate the brain’s collocation of storage and processing. At the heart of this revolution is the memristor (RRAM). By leveraging Ohm’s Law and Kirchhoff’s Current Law for analog computation, memristive crossbar arrays offer massive parallelism and energy efficiency superior to digital CMOS.

This report presents a state-of-the-art review of memristors and resistive switching [2], tailored for circuit designers.

II. HISTORICAL EVOLUTION: FROM THEORY TO SILICON

The history of the memristor is a narrative comprised of two intertwined timelines: the experimental observation of resistive switching in oxides, dating back to the 1960s, and the theoretical derivation of the “fourth fundamental missing circuit element” in the 1970s. Understanding this duality is essential for grasping the current terminology and the engineering challenges that persist today.

At the device level, memristors exhibit strongly nonlinear switching behavior governed by coupled physical processes such as ion migration, filament formation, and thermal effects. These dynamics are typically modeled using stiff ordinary or differential–algebraic equations, where relevant time scales can differ by several orders of magnitude. At the circuit level, large crossbar arrays introduce additional challenges, including parasitic resistances, sneak paths, and IR drop, which significantly affect computational accuracy and convergence. Accurately capturing these effects is essential for evaluating real-world performance and reliability.

A. Professor Leon Chua’s ‘the Missing element’

Chua noted that there are six different mathematical relations connecting pairs of the four fundamental circuit variables: electric current i , voltage v , charge q and magnetic flux ϕ .

Thus, there should be four basic circuit elements described by the remaining relations between the variables.

The final one to complete the four elements being the memristor defined by a certain memresistance M provides a functional relation between charge and flux. A defining unique characteristic of the memristor is it offers frequency-dependent Lissajous figure, which can’t be reproduced by any other nonlinear component. Such that

$$v = R(w)i$$

Chua and Kang generalized the memristor concept to a much broader class of nonlinear dynamical systems they called memristive systems

$$v = R(w, i)i$$

$$\frac{dw}{dt} = f(w, i)$$

B. Experiments

Parallel to Chua’s theoretical work, experimental physicists were observing “anomalous” switching behaviors in thin dielectric films. As early as 1962, Hickmott reported low-frequency negative resistance in anodic oxide films, including aluminum oxide and silicon oxide. Throughout the 1960s and 70s, researchers observed that applying high voltage pulses to oxides like Nickel Oxide (NiO) or Niobium Oxide (Nb_2O_5) could reversibly switch them between high and low resistance states.

C. 2008 revolutionary Unification

The watershed moment for the field arrived in 2008. A team at HP Labs, led by R. Stanley Williams and including Dmitri Strukov, published “The missing memristor found” in Nature [3]. The team had been investigating crossbar switches made of Titanium Dioxide (TiO_2) and observed coupled electronic and ionic transport. They realized that the drifting of oxygen vacancies (dopants) under an electric field caused a change in the resistance that perfectly matched the mathematical formalism of Chua’s 1971 memristor.

The HP paper [4] proposed a simple physical model: the device consists of a semiconductor film sandwiched between two metal electrodes oxides. This linked the abstract circuit theory of 1971 with concrete nanoscale physics, provided a unifying vocabulary for the field.

the term “memristor” became the de facto nomenclature in the neuromorphic community for any two-terminal device with non-volatile memory.

D. Neural Networks and Neuromorphics

Following the 2008 discovery, the focus shifted from verification to integration. In 2010, HP Labs demonstrated that memristors could perform boolean logic (stateful logic) and implemented within chips. “*HP has created development-ready architectures for memory chips using memristors and believes it is possible that devices incorporating the element could come to market within the next few years.*” [4], implying that memory arrays could inherently compute. By 2015, the industry began major commercial moves. Intel and Micron announced 3D XPoint technology, a resistive memory class (likely Phase Change Memory, PCM, or a variant of RRAM) that promised speeds closer to DRAM with the non-volatility of Flash.

Simultaneously, the rise of Deep Learning created a voracious appetite for matrix algebra computation. Groups from Stanford, IBM, and Politecnico di Milano demonstrated how memristor crossbars would be quintessentially necessary for naturally performed Matrix-Vector Multiplication (MVM) [5].

III. RESISTIVE SWITCHING MECHANISMS & DEVICE PHYSICS

As described earlier, these RRAM systems rely on active metal electrodes and a counter-electrode (e.g., Platinum, Pt, or Tungsten, W).

Resistive switching devices operate by toggling between two distinct resistance states: Low Resistance State (LRS) and High Resistance State (HRS).

SET Process: When a positive bias is applied to the active electrode, metal atoms oxidize into cations (Ag^+ , Cu^{2+}). These ions migrate through the solid electrolyte layer (such as amorphous Silicon or a chalcogenide) toward the inert cathode. Upon reaching the cathode, they are reduced back to metallic atoms, nucleating a filament that grows back toward the anode. When the filament bridges the gap, the device switches to a Low Resistance State (LRS).

RESET Process: Reversing the polarity dissolves the filament via electrochemical oxidation, retracting the bridge and restoring the High Resistance State (HRS).

We can also distinguish two different rules/modes of switching

1) *Unipolar vs. Bipolar Switching Rules:* Bipolar Switching: Requires voltage of opposite polarity to SET and RESET. This is the standard for ECM and VCM devices and aligns naturally with biological potentiation and depression.

Unipolar Switching: Switching depends only on voltage magnitude and pulse width, not polarity. Typically, a short high-voltage pulse causes Joule heating to fuse the filament (SET), while a longer, lower-voltage pulse causes thermal dissolution (RESET). Unipolar switching is less common in modern neuromorphic designs due to higher power consumption and the complexity of controlling the thermal reset.

The simulation result for the unipolar memristor switching behavior is illustrated in Fig. 7b, showing that the memristance of the device can increase or decrease for the same voltage polarity. [6].

IV. CROSSBAR ARCHITECTURES: THE ANALOG CORE

1) *Neuromorphic Hardware:* Mimicking the neural structure of the brain, an energy efficient and adapted topology for on-hardware machine-learning model can be implemented. Yet to achieve an efficient execution of Spike neural networks for instance, utilization of individual neuromorphic cores and minimizing inter-core communication must be done, hence why different topologies exist for resistive switching devices within neuromorphic structures. [7]

The most common configuration is the 2-terminal crossbar array, which uses a grid of word lines (WLs) and bit lines (BLs) to form programmable nodes at each intersection, mimicking artificial synapses.

The physical arrangement of memristors determines the density, efficiency, and feasibility of the computing system.

Here are the main structures to take into consideration for such structures.

2) *Passive Crossbar Arrays (OTIR):* This offers the highest possible integration density and is suitable for 3D stacking (Vertical RRAM or VRRAM), where multiple layers of crossbars are built on top of CMOS logic.

The Sneak Path Problem [8]. This is the fatal flaw of passive arrays. When attempting to read a specific high-resistance cell (Logic ‘0’), current can bypass the selected cell and flow through parallel paths formed by surrounding low-resistance cells.

V. CIRCUIT-LEVEL SIMULATION & ANALYSIS

A. Model of memristor

As memristors can’t be found in the standard library. Creating a subcircuit must be done first (.subckt).

To simulate the resistive switching behavior, we utilized the SPICE model developed by Biolek et al. [9], though other models like VTEAM [10] are also widely used. This model extends the linear ion drift model originally proposed by HP Labs by incorporating a nonlinear window function (the ‘Biolek

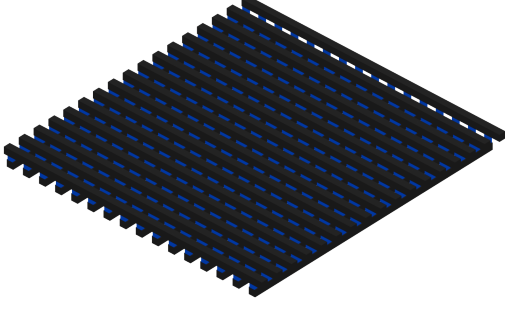


Fig. 1: Passive Crossbar Array Simplified Model

Window') to resolve boundary issues at the memristor's state limits ($x = 0$ and $x = 1$).

The VTEAM model uses a threshold-based model ($v > v_{on}$) that is adaptable to any switching mechanism (linear, exponential, or tunneling).

We selected the Biolek model for this study due to its numerical stability in crossbar grid simulations, while acknowledging that VTEAM offers superior physical accuracy for single-device characterization.

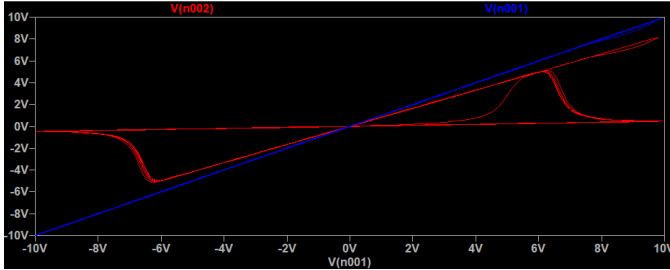


Fig. 2: Memristor Biolek Model I-V characteristic

B. Passive Crossbar Array (OTIR) Simulation

To investigate the feasibility of memristive crossbar arrays for analog computing, we constructed a 3×3 passive crossbar array using the Biolek memristor model in LTspice (Fig. 3).

As described earlier the passive crossbar OTIR circuit consists of three distinct functional blocks,

- Input Layer (Rows), represented by voltage sources V_1, V_2, V_3 , which encode the input vector data as analog voltage levels (e.g., $0.5V$ for logic '1' and $0V$ for logic '0').
- Weight Matrix (Memristors), the memristors are placed at each crosspoint. Their conductance state ($G = 1/R$) represents the stored "weight" of the neural connection. Low Resistance States (LRS, $1k\Omega$) represent high correlation (logic '1'), while High Resistance States (HRS, $100k\Omega$) represent low correlation (logic '0').
- Output Layer (Columns), where the output is the result of effectively summing the currents from all connected rows according to Kirchhoff's Current Law (KCL).

We decided to configure the array according to a Single-Layer Perceptron (SLP) designed to recognize a specific input

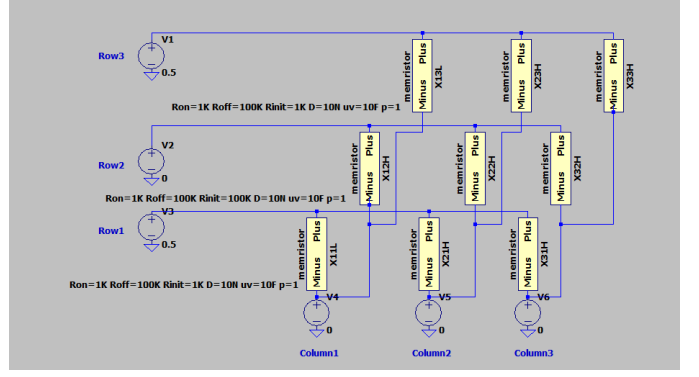


Fig. 3: Passive Crossbar Array Simplified Model

pattern in the 3×3 array of memristors. The memristors in Column 1 were programmed to match the input vector, while Columns 2 and 3 were programmed with mismatching weights to serve as controls.

Looking at the the initial value of resistance or R_{init} you can determine how each memristor was configured.

Then an input voltage of $[0.5V, 0, 0.5V]$ was applied to the rows.

Column 1 was configured with a High Resistance State ($100k\Omega$) at the intersection of Row 2.

Conversely, the mismatched columns resulted in a current of only $5\mu A$, providing a clear On/Off ratio of 100:1. This demonstrates that the array can successfully distinguish between matching and non-matching patterns. By applying a threshold (e.g., $I_{thresh} = 250\mu A$)

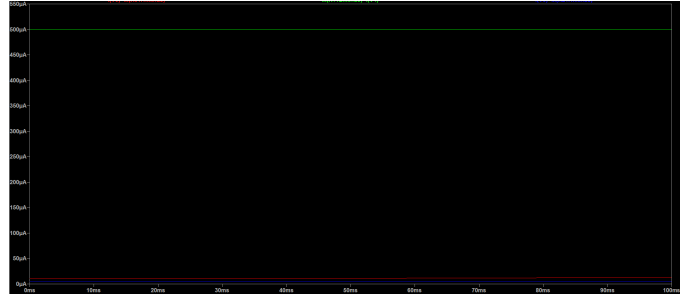


Fig. 4: Transient simulation results showing the output current response. Column 1 (Green) exhibits a "Match" current of $500\mu A$, while adjacent columns (Red/Blue) show negligible leakage current ($5\mu A$).

While this 3×3 simulation validates the fundamental physics, scaling this architecture to commercial neuromorphic systems introduces specific challenges and opportunities.

The single column demonstrated here acts as a single neuron. In a large-scale system, thousands of columns are used in parallel. A matrix of size $N \times M$ can perform an entire layer of a Deep Neural Network (DNN) in a single clock cycle. This theoretical throughput assumes negligible parasitic delays and ideal interconnects.

To accurately measure the accumulated column current without disrupting the voltage potential across the memristors, we implemented a Transimpedance Amplifier (TIA) stage. Unlike a simple shunt resistor, which introduces a voltage burden that reduces the effective drive voltage across the

synaptic devices, the TIA maintains a Virtual Ground at the column node

C. Active Arrays: One-Transistor-One-Resistor (1T1R)

While passive (1R) arrays offer density, they suffer from “sneak path” currents in larger grids (e.g., 64×64), where leakage through neighboring cells degrades accuracy. The industry standard solution, adopted by TSMC and Intel, is the 1T1R topology.

In 1T1R, a selector transistor is added to every memristor. Acting as a switch, the transistor isolates the cell when OFF, eliminating sneak paths and enabling massive arrays limited only by interconnect resistance.

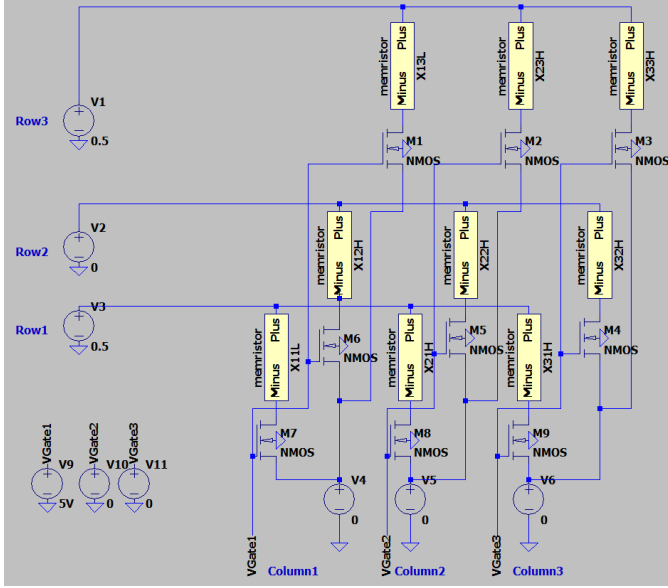


Fig. 5: Active 3×3 Array: One-Transistor-One-Resistor (1T1R) SPICE Circuit

The introduction of the 1T1R architecture introduced a series resistance component intrinsic to the selector transistor. While the passive (1R) array achieved a theoretical maximum current of $500\mu A$ ($R_{on} = 1k\Omega$), the 1T1R cell limited this current to $44\mu A$ due to the channel resistance ($R_{ds,on} \approx 10k\Omega$) of the NMOS selector. This 10x reduction in current highlights the trade-off between the ideal linearity of passive arrays and the necessary selectivity of active arrays.

Although this reduces the absolute signal magnitude, it advantageously limits the total power dissipation of the array. To compensate for this voltage divider effect, readout circuits (TIAs) in 1T1R systems typically require higher gain stages compared to passive crossbar implementations.

On the other hand this gives far more control; by modulating the gate voltage of the selector transistor, the driver sets a hard limit on the current that can flow through the memristor during SET.

VI. ADVANCED COMPUTATIONAL CHALLENGES

The analog nature of memristive computation introduces challenges noise, offsets, non-linearities, and parasitic voltage drops.

In a neuromorphic core, the “Read” operation is often a Matrix-Vector Multiplication.

A. Analog Matrix-Vector Multiplication (MVM) Deep Dive

The memristive crossbar performs Matrix-Vector Multiplication (MVM) by exploiting two fundamental physical laws in a single computational step.

Multiplication: At each crosspoint (i, j) , the current flowing through the memristor is the product of the applied row voltage and the device conductance,

$$I_{ij} = V_{in,i} \cdot G_{ij}$$

where $G_{ij} = 1/R_{ij}$ represents the programmed synaptic weight.

Summation: At each column (bitline), all currents from connected rows sum together,

$$I_{out,j} = \sum_{i=1}^N V_{in,i} \cdot G_{ij}$$

This is precisely the definition of a matrix-vector product: $\mathbf{I}_{out} = \mathbf{G}^T \cdot \mathbf{V}_{in}$. The crossbar thus computes an entire column of a weight matrix in $O(1)$ time complexity, compared to $O(N)$ for sequential digital multiplication.

Neural network weights $w \in [-1, 1]$ must be mapped to physical conductances $G \in [G_{off}, G_{on}]$.

B. Mode of Sensing

There are also two main means of sensing, current-sensing and voltage-sensing. As I described earlier a TIA would ideally be used, but as it consumes significant area and static power a more compact “Class AB” or inverter-based TIAs is key. On the other hand voltage switching relies on charging the bit line and using a comparator to read the output.

C. IR Drop Problem

In large crossbar arrays (e.g., 128×128 or larger), the metal wires (Word Lines and Bit Lines) have non-zero resistance (R_{wire}). As current flows through the cells, voltage is lost along the wire.

A memristor intended to receive $0.5V$ might only receive $0.42V$, causing a non-linear error in the multiplication result ($I \neq G \cdot V_{target}$). This effectively limits the maximum size of the crossbar before signal integrity degrades below detectable levels.

Analog computing fundamentally trades precision for energy. Because memristive inference relies on detecting precise current levels, noise generated by the random motion of electrons in the resistive filament, could introduces a Gaussian error to the summation current.

Furthermore linearity might not work because, unlike an ideal resistor following Ohm’s Law ($I \propto V$). Real memristors, exhibit non-linear I-V characteristics.

We discussed on how these structures may help accelerate MVM operations, but on the downside, because the input has to be converted with a DAC to analog, and back to digital with an ADC. These operations would consume the majority of the power to fulfill these operations and remains the main obstacle preventing memristors from replacing standard RAM.

D. Neuromorphic Architectures

1) *Analog Matrix-Vector Multiplication (MVM) Engines:* This approach aims to accelerate standard Deep Neural Networks (DNNs) used in AI today.

The heavy lifting in DNNs are MAC operation. A memristor crossbar performs this physically in one step, regardless of the matrix size

2) *Spiking Neural Networks (SNNs) and STDP:* SNNs communicate via discrete spikes rather than continuous values, more closely mimicking biological neurons. Memristive synapses naturally implement Spike-Timing-Dependent Plasticity (STDP), where the weight update depends on the relative timing between pre- and post-synaptic spikes. The memristor's analog state evolution under voltage pulses directly realizes STDP learning rules without explicit digital computation.

3) *Edge AI and Chiplet Integration:* The convergence of memristive computing with edge AI presents compelling opportunities: ultra-low power inference through analog MVM, on-chip learning via STDP-based adaptation, and direct analog processing of sensor inputs. Looking toward the late 2020s, heterogeneous “Chiplet” architectures—placing analog RRAM cores alongside high-performance digital logic—promise to unlock the next generation of AI performance at the edge.

VII. SCALABILITY & BENCHMARKING

A. IR Drop Investigation

In large crossbar arrays, finite wire resistance causes progressive voltage attenuation along wordlines and bitlines. For a wire segment resistance R_{wire} , the effective voltage at crosspoint (i, j) becomes,

$$V_{eff}(i, j) = V_{source} - I_{cumulative} \cdot R_{wire} \cdot j$$

We analyzed IR drop impact using Modified Nodal Analysis (MNA) with sparse matrices, solving the full resistor network. Fig. 6 shows the voltage attenuation across a 32×32 array with $R_{wire} = 10\Omega$.

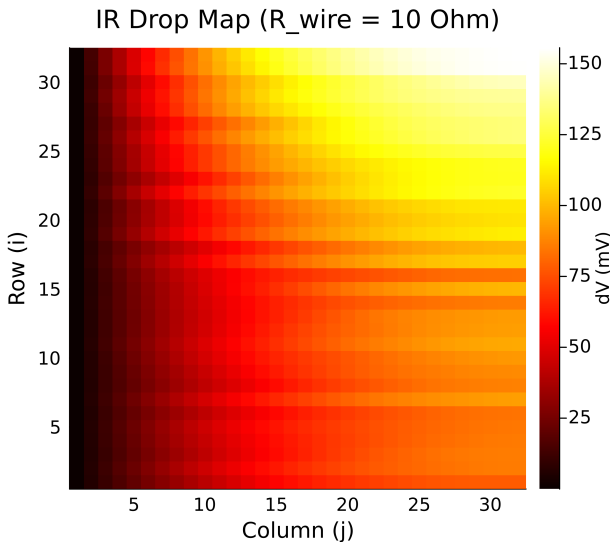


Fig. 6: IR Drop heatmap showing voltage attenuation from source (top-left) to far corner (bottom-right) in a 32×32 crossbar.

Table I: IR Drop impact on MVM accuracy for a 32×32 array

Wire Resistance	Relative MVM Error
$R_{wire} = 0\Omega$	0% (reference)
$R_{wire} = 1\Omega$	$\sim 2\%$
$R_{wire} = 10\Omega$	$\sim 8\%$
$R_{wire} = 50\Omega$	$\sim 20\%$

Additionally 64×64 is relatively small, and the error gets even bigger the larger the array.

This voltage degradation introduces non-linear error in the multiplication result ($I \neq G \cdot V_{target}$), effectively limiting maximum crossbar size.

B. Large-Scale Simulation Framework with Julia

While LTspice is effective for small-scale validation (3×3 arrays), it becomes computationally intractable for arrays exceeding 64×64 . To address this scalability limitation, a Julia-based simulation framework, was used leveraging **DifferentialEquations.jl**, **SparseArrays.jl**, and **ForwardDiff.jl**.

1) *Biolek vs. VTEAM Model Comparison:* We implemented two industry-standard memristor models to compare their behavior:

Biolek Model [9]: Which extends the original HP linear ion drift model by introducing a non-linear Window Function. In physical devices, the conductive filament cannot grow beyond the device thickness or shrink below zero,

$$f(w) = 1 - \left(\frac{w - w_{on}}{w_{off} - w_{on}} \right)^{2p}$$

VTEAM Model [10]: The Voltage ThrEshold Adaptive Memristor model uses threshold-based switching with non-linearity exponents, designed to fit experimental data from specific materials. Below this voltage, the ions do not migrate,

$$\frac{dw}{dt} = \begin{cases} k_{off} \cdot \left(\frac{v}{v_{off}} - 1 \right)^{\alpha_{off}} \cdot f(w) & v < v_{off} \\ 0 & v_{off} \leq v \leq v_{on} \\ k_{on} \cdot \left(\frac{v}{v_{on}} - 1 \right)^{\alpha_{on}} \cdot f(w) & v > v_{on} \end{cases}$$

VTEAM offers superior physical accuracy for single-device characterization due to its threshold-based formulation, while the Biolek threshold model provides better numerical stability in large crossbar simulations, hence the Biolek model was selected to ensure robust convergence in SPICE.

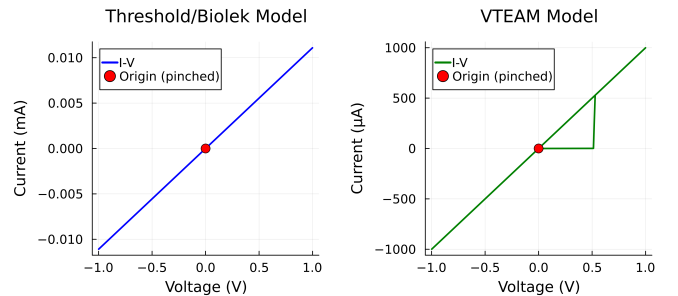


Fig. 7: I-V hysteresis comparison between Threshold/Biolek (blue) and VTEAM (green) models under identical triangular voltage stimulus.

2) *Scalability Benchmarks*: Table II shows simulation timing for ideal MVM and full IR drop analysis across array sizes.

Table II: Julia simulation timing benchmarks

Array Size	Ideal MVM	IR Drop (MNA)
32×32	0.1 ms	1 ms
64×64	0.4 ms	4 ms
128×128	2 ms	20 ms
256×256	10 ms	100 ms
512×512	50 ms	~ 1 s

Julia handles 512×512 arrays in approximately 1 second—simulations that would require hours in traditional SPICE. Julia can be used to determine the conductance states for the memristors.

3) *Non-Ideality Error Analysis*: We quantified the impact of device variability and read noise on MVM accuracy:

Table III: Error contribution from device non-idealities

Configuration	Relative Error
5% device variability	$\sim 5\%$
5% variability + 2% noise	$\sim 7\%$

C. Differentiable Simulation for Hardware-Aware Training

A unique advantage of Julia’s ecosystem is automatic differentiation through ForwardDiff.jl, enabling gradient-based optimization of device parameters. By defining a differentiable simulation function, we can compute,

$$\frac{\partial w_{final}}{\partial \theta}$$

where θ represents any kind of device parameters (e.g., switching rate k_{on} , threshold voltage V_{th}).

This capability enables, backpropagating through the physical device model and optimizing device parameters for target conductance states.

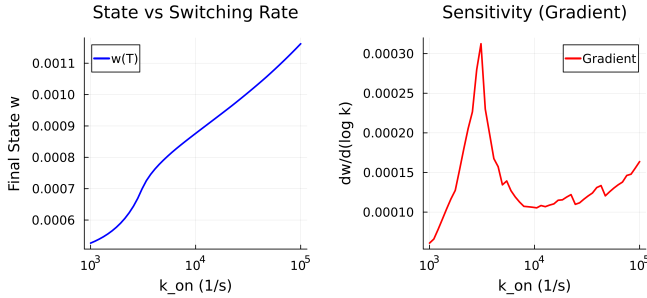


Fig. 8: Parameter sensitivity landscape showing final memristor state as a function of switching rate k_{on} , with gradient computed via automatic differentiation.

This differentiable approach bridges the gap between abstract neural network training and physical device constraints.

D. Future Work: Xyce Integration

For SPICE-accurate validation of large arrays, integration with Sandia’s Xyce simulator provides a path forward. Xyce offers parallel SPICE simulation with Verilog-A support, enabling physically precise memristor models while maintaining Julia’s control and analysis capabilities. This hybrid workflow—Julia for rapid exploration, Xyce for final validation—represents the future of memristive circuit design.

VIII. RELIABILITY & DEVICE CHALLENGES

No two memristors are the same.

Manufacturing variations cause the “SET” and “RESET” voltages to vary across the chip. A pulse that successfully writes to Memristor A might fail to write to Memristor B, requiring complex “Write-Verify” closed-loop controllers.

Additionally endurance is also a major concern, Memristors wear out. The conductive filament physically breaks and reforms. After 10^6 to 10^9 cycles, the device may get stuck in a permanent Low or High resistance state.

Finally, energy efficiency is a key driver. As noted in [11], data movement in classic architectures consumes $\sim 100\times$ more energy than computation itself, a penalty that in-memory memristive architectures aim to eliminate.

IX. CONCLUSION

The transformation of the memristor from a theoretical construct to a commercial technology bridges the gap between the “Memory Wall” and the “Power Wall.” The frontier has now shifted to circuit and system-level optimization. Successful deployment relies on innovations in current-mode sensing and adaptive write-verify schemes.

Looking toward the late 2020s, integrating crossbar arrays into heterogeneous “Chiplet” architectures—placing analog RRAM cores alongside digital logic—promises to unlock the next generation of AI performance, bringing brain-like efficiency to the edge.

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